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An overview and comparison of the Versatile Video
Coding standard

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Abstract

This thesis mainly focuses on the Versatile Video Coding standard. It goes over the history of relevant video coding technologies of the same nature, giving a brief rundown of significant details of each piece of technology, with illustrations throughout. It tells about the development of Versatile Video Coding and relevant context, ending that section with a report about industry adoption of this technology and some future projections. It goes into detail about specific improvements upon preceding technology, alongside all relevant context thereof. In the methods section, it describes ways in which metrics of the quality of video footage can be objectively measured, calculated and compared. The next section presents the reader with results of objective video testing performed.

Measured data suggests that the VVC codec is generally superior to prior video coding technology in quality. Available data seems to hint at VVC seeing wider adoption in future, despite competition. A window into performance aspects of this technology, that this thesis gives, should inform the author of those aspects as well, and help them choose between some of the presets VVC works on.

Keywords

Video coding, Video compression, Standardization

Abstrakt

Tato práce se v první řadě zabývá video kódovacím standardem Versatile Video Coding. Zmiňuje historii relevantních video kódovacích technologií stejného charakteru, u kterých do mělka prochází jejich významnější detaily, a graficky ilustruje některé koncepty. Dále se zabývá historií vývoje Versatile Video Coding a relevantního kontextu, a zakončuje shrnutím prozatímní adopce VVC průmyslem. V další části zmiňuje způsoby, jimiž VVC zdokonaluje předchozí technologie, a kontext v kterém tomu činí. Dále jsou v metodice představeny některé metody, pomocí nichž lze objektivně měřit a porovnávat kvalitu video sekvencí. V další části představuje výsledky měření kvality videa podle zmíněných metod.

Naměřená data vypovídají o tom, že VVC obecně dosahuje znatelně lepších výsledků kvality, než předchozí video kódovací technologie. Dostupné informace napovídají, že VVC uvidí širší adopci průmyslem, i přes silnou konkurenci. Práce dává čtenáři i pojem o výkonových stránkách, na základě nichž by se mohl rozhodovat i např. mezi presety s kterými VVC přichází.

Klíčová slova

Kódování videa, Komprimace videa, Standardizace

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List of Abbreviations

AVC Advanced Video Coding

CFE Call for evidence

CFP Call for proposals

DCT Discrete cosine transform

DPCM Differential pulse-code modulation

GOP Group of pictures

HD High definition

HDR High dynamic range

HEVC High Efficiency Video Coding

HLG Hybrid log gamma

JVET Joint Video Exploration Team (2015-2017), Joint Video Experts Team (2017-)

MPEG Moving picture experts group

NTSC North American standard for television

PAL Phase alternating line

PQ Perceptual quantization

SAO Sample adaptive offset

SD Standard definition

UHD Ultra-high definition

VCEG Video coding experts group

VVC Versatile Video Coding

WCG Wide color gamut

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Introduction

Video coding technology is another one of these things that a big part of society depends on and uses daily, yet an understanding of it, that isn't shallow, is far less common. Ever since the introduction of Digital broadcasting, this technology has only been increasing in its use whether it's for news, entertainment, education, or anything else. Along with developments in digital technology as a whole, video today is able to be used in qualities incomparable to what they were just some decades ago. The introduction and mass adoption of the internet only boosted the use of this technology and created more incentive to innovate.

In 2022, 65% of internet traffic was made up of video data [38], up from 51% in 2016 [64]. Adjusted for total internet traffic, however, this is almost a quadrupling of traffic made up of video [10]. This could be due to increased use in general, as well as streaming at higher bitrates. The covid-19 pandemic has also had a major effect on those metrics, especially via the use of video conferencing, use of those technologies grew by 62% during the pandemic [56].

This is in a big way caused by the disproportionate amount of data that has to be transmitted to facilitate video technology. To compare, images usually don't go past lower double digit MB, a minute of audio will, depending on its quality, usually stay under about 25 MB, while a minute of video in HD is usually not under 100 MB, and can go way past that.[77][82][57]

This puts pressure on the industry to figure out ways to compress video into smaller file sizes. One of the most basic, yet foundational of ideas was patented by Ray Davis Kell in 1929, he wrote *"It has been customary in the past to transmit successive complete images of the transmitted picture. [...] In accordance with this invention, this difficulty is avoided by transmitting only the difference between successive images of the object."*. This idea took a long time to actually be used in practice, however, it's still a cornerstone of most video standards today [52].

This thesis will mainly focus on the, as of the writing newest, Versatile Video Coding standard introduced in 2020. The goal is to objectively compare the quality of the technology, to previous technologies of similar characteristics.

Goals

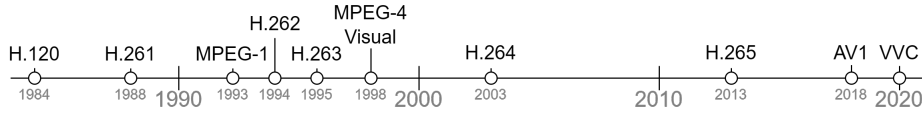
The main goal of this thesis is to explore the Versatile Video Coding standard, provide relevant context and measure objective aspects of the technology. This is supported by the following sub-goals:

- Put VVC into historical context within the field,
- Go over the process of development of VVC,
- Explain specific improvements it makes while putting them into context,
- Objectively measure quality and performance aspects, using particular methods.

1 A brief history of video coding technology

Video coding standardization has a colorful history. As an integral part of information and communication technology as a whole, it's always been influenced by the field status quo, and made an effort to adapt, whether it's the inclusion of interlacing, adoption of HDR video or optimizing for network throughput. This section will mainly go over the history of the H.26x series of video coding standards, with detours to mention other influential ones on the outside.

Figure 1: Timeline of video coding format releases. Note: chart created by author from data retrieved from Historical timeline of video coding standards and formats (Iain Richardson, 2024).



1.1 H.120

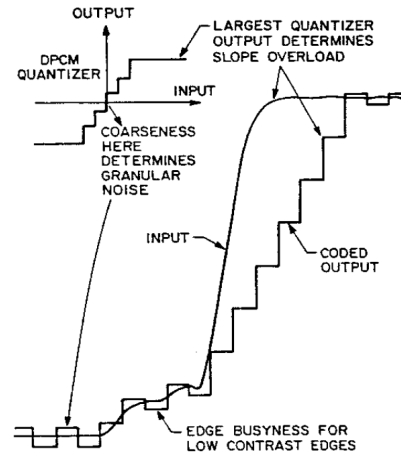
As the first video coding standard, H.120 was finalized in 1984. It was developed by the COST 211 European research project, and published by the CCITT (Comité Consultatif International Téléphonique et Télégraphique), which was later renamed to ITU-T (International Telecommunication Union Telecommunication Standardization Sector) - the organization behind the creation of many future coding standards.[35][14]

H.120 streams ran at 1544 kbit s^{-1} for NTSC (North american standard for television) and 2048 kbit s^{-1} for PAL (Phase alternating line). It featured several novel techniques for the purpose of better compression of the considerably data-heavy video compression and transmission.[35]

Among the techniques featured was conditional replenishment. [35] It's an interframe compression method that uses data correlation in consecutive frames to reduce the amount of information that has to be transmitted. It works by detecting which portions of a frame have changed since the last one, and only sending those parts. The receiving side then uses this data, plus the data from the last frame, to construct back the full picture. The amount of compression that can be achieved through this technique is significant, however, it fully depends on the amount of change between the frames. The data compression rates usually go from 10 to 1 to about 2 to 1.[40]

Another technique featured is Differential Pulse Code Modulation (DPCM). At the start, a video is a set of frames, each of which consists of pixels with color information, represented by multiple numerical values in certain ranges. DPCM achieves compression by, instead of always transmitting those values in their full range, sending only the amounts the value has changed from the last frame. Due to these change values usually being significantly smaller, a reduced range can be used, lessening the amount of information needed to be sent. That's not to say this is flawless. DPCM can introduce video artifacts, depending on how restrictive the new range is. Problems such as edge busyness, granular noise or slope overload have been described (Figure 2). Differences between the two curves represent deviations of the transmitted data to the original video data.[37]

Figure 2: DPCM signal distortions. Source: [37]DPCM signal distortions. Retrieved from DPCM - Overview (Bernd Girod, 2000)



Even after achieving significant savings in data, it was clear that this wasn't enough, and more advanced coding would have to be developed.

1.2 H.261

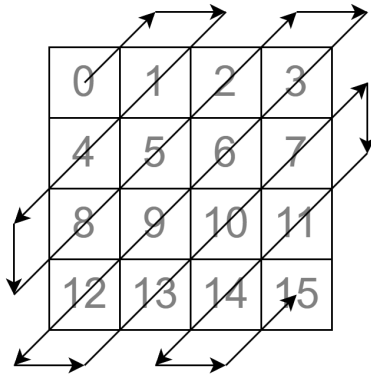
This standard was approved by CCITT in 1988. It has been the first of a line of truly widely adopted codecs, and has served as a foundation for them in its features and technologies. As a video codec, it uses lossy compression to compress video files while main [18]

To reiterate, H.261 introduced several new coding techniques, which would define a large part of future standardization efforts, as all subsequent international video coding standards in the H.26x series have been based closely on the H.261 design. Additionally, this codec provided a, at the time, much needed benchmark for video codec performance and capability.[18]

This standard was the first one, where the concept of a macroblock has appeared. It's the basic processing unit of the design. H.261's macroblocks consist of a 16×16 array of luma (or luminance) samples, and two half-resolution arrays of chroma (or chrominance) samples, at 8×8 . As implied, the YCbCr color space is used - a color space where a color is represented by a luminance value - basically a brightness (represented by Y), and two

chrominance values, Cb and Cr, named blue-difference and red-difference. This format benefits from the ability to represent a lot of detail in the Y value, giving it more resolution, and coloring the image in with chroma values for which a lower resolution can suffice. [76][45][8] This standard also uses the zig-zag scanning pattern (Figure 3).

Figure 3: Zig-zag scanning pattern, simplified on a 4x4 grid. Note: diagram created by author.



Perhaps the most significant technique utilized by this standard is Discrete Cosine Transform (DCT). First proposed by Nasir Ahmed in 1972, DCT is a way of representation of a set of data points as a sum of cosine functions of different frequencies, by specifying weights that each of the functions is to be multiplied by in the calculation, based on the source data, here, based on the values of color components in a given image. DCT also allows for the parametrization of the resolutions that can be used to store the weights, effectively controlling the quality in which the source images are portrayed by this technique. This technique allows for lossy compression of significant

rates with little visual difference. Usually, the most affected areas of the images are those that feature a lot of small-scale detail - a lot of differences between pixel values in small areas. Those are, however, uncharacteristic for most common video sources.[67][20]

In figure 4, all available combinations of cosine waves with different frequencies, but constant weights, can be seen. Each of them fills an 8×8 grid of pixels. Upon compression, each 8×8 block of pixels is assigned a certain combination, to which weights are assigned based on the data.

1.3 MPEG-1

The MPEG-1 standard, released in 1993, was designed to achieve acceptable video quality at 1.5 Mbit s^{-1} . [52] It achieves audio compression of 6:1 and video compression of 26:1, without excessive quality loss [73]. It has been developed with, at the time, common storage technology, such as CD-ROM, digital audio tapes, tape drives and several others [34]. A thing to note is that it, for the first time, introduced the *MPEG-1 Audio Layer III* or MP3 audio format - a format well-known to this day [39].

The main concern with this format, and the main motivation for the development of MPEG-2, was the lack of support for interlaced video [52]. Interlaced video is a technology that is mainly used in TV devices. It allows for the perceptual doubling of framerate at no extra bandwidth cost [36]. It works by only updating all odd or all even rows at a time, which results in a smoother viewing experience for the human eye. While quite effective, it can produce artifacts, especially upon fast motion in the video. [59]

1.4 MPEG-2/H.262

MPEG-2 was specified in 1994 as ISO/IEC 13818 and as ITU-T recommendation H.262. It was the first joint development by Video Coding Experts Group (VCEG) and Motion Picture Experts Group (MPEG) [44]. It supports both standard definition (SD) video and high-definition (HD) video. The standard was meant to deliver studio-quality digital video at rates of 3 Mbit s^{-1} - 10 Mbit s^{-1} for SD, and was designed with quality vs. complexity trade-offs. [52]

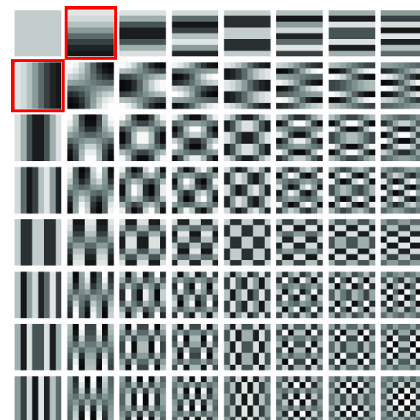
All standard-compliant MPEG-2 video decoders are backwards compatible with MPEG-1, despite its lack of support for interlaced video.[86]

1.5 H.263

Finished in 1995, only about a year apart from H.262, H.263 was a big step forward in video coding. The main aim in designing this standard was enabling video conferencing at low bit rates for mobile wireless communications. Especially for progressive (non-interlaced) video, the quality was superior to all prior standards at all bit rates. [52]

The first version of the standard supported 5 distinct picture magnitudes, named Sub-QCIF, QCIF, CIF, 4CIF, 16CIF in order of size, with Sub-QCIF having 128×96 luminance samples at the low-end, and 16CIF having 1408×1152 of those samples. Chrominance is, as is customary, halved in resolution. [46]

Figure 4: Two-dimensional DCT frequencies. Retrieved from Image Compression Based on a Partially Rotated Discrete Cosine Transform With a Principal Orientation (Lee, Gihwan and Choe, Yoonsik, 2021)



H.263 is used in H.324, H.323 and H.320 video conferencing standards, as well as in web-distributed video content that plays with Macromedia's Flash 7 player. The 3GPP cell phone standardization project includes the H.263 video codec for transmission of video to and from mobile phones. [52]

H.263, unlike most codecs, defines fundamentals of an encoded video in terms of the maximum bit rate per frame (picture), or "BPPmaxKb". During encoding, a value is selected for "BPPmaxKb", and then the video cannot exceed this value for each frame. The final bit rate, the frame rate, the compression, the chosen resolution and block format will all depend on this. As a video conferencing-focused codec, performing this process aims to provide a smooth transmission, allowing for uninterrupted conversations, at the cost of occasional drops in momentary quality. [85]

1.6 AVC/H.264

Advanced video coding (AVC), also known as MPEG-4 Part 10, and standardized by ITU-T as H.264, was created in 2003 and first published the following year. It follows in its predecessors' footsteps in most features, however, its compression capability is significantly better. The design goal was to create video coding standard capable of a compression rate twice that of preceding standards, with as little loss of visual quality as possible. [52]

AVC is a highly flexible codec. It features several profiles varying in capability. For example, profiles vary in how optimizations are applied, some minimizing bandwidth requirements, some favoring quality as the main goal. Profiles also vary in color format, using either 8-bit or 10-bit color, and in chroma subsampling ratios. [85]

This standard is highly used to this day, e.g. YouTube uses it as the default standard for most videos. As mentioned, it is also one of the coding standards used for Blu-Ray video, along with VC-9. [4][58]

1.7 HEVC/H.265

High Efficiency Video Coding (HEVC), also known as MPEG-H Part 2 and standardized as H.265 by ITU-T, was developed jointly by MPEG of ISO/IEC and VCEG of ITU-T, forming the Joint Collaborative Team on Video Coding - JCT-VC. HEVC was developed and approved in 2013, with the express goal of increasing video compression rates to much higher than that of AVC while preserving video quality, as technologies such as Ultra HD television were being brought into use. At the time, Microsoft stated that their new OS, Windows 10, would feature a HEVC decoder. For transmission between physical media, the HDMI connector suffices, supporting

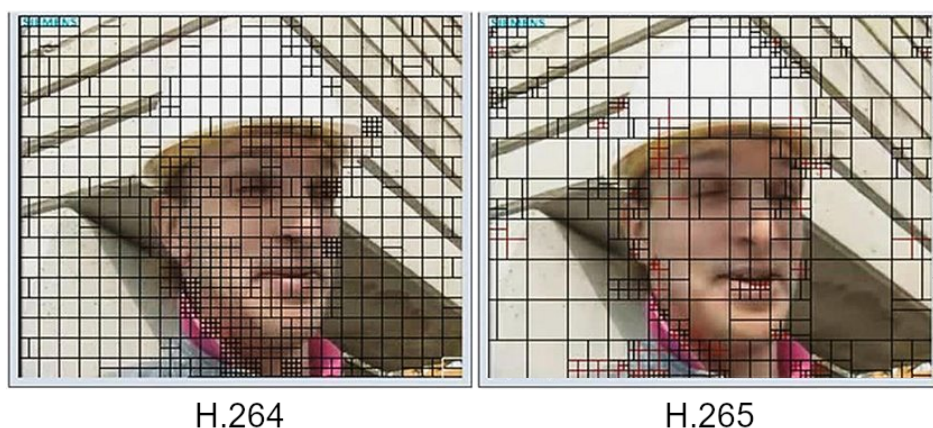
UHD at 60Hz, however, the DisplayPort connector is more suitable, providing support for resolutions as high as 8K. Mobile High-definition Link (MHL) is also among the suitable connectors, mainly used to interconnect mobile devices, but not limited to this use. [69]

Another advancement of this standard is its adaptation to modern processors. The structure of video coding and decoding processes, as well as organization in memory, was specially designed to support efficient parallelization using multiple cores of a CPU.[85]

HEVC also brings advances in the usage of frame divisions into blocks. Instead of statically sized macroblocks, that formats such as MPEG-2 and JPEG use, HEVC introduces the Coding Tree Unit (CTU) allowing for scaling based on level of detail in certain areas. While macroblocks are typically made up of 16×16 samples, CTUs can be between 16×16 and 64×64 samples in size (allowing for sizes being powers of 2). CTUs can break down further into Coding blocks (CBs), depending on the level of detail in a given area of a picture. This allows for significant data and coding efficiency savings in most video streams. [89][69]

In figure 5, a comparison of how a video stream may be subdivided by the H.264 and H.265 video coding standards can be seen. As the picture shows, H.265 uses blocks that can be a lot bigger, especially in more flat areas, those with sparse detail.

Figure 5: Comparing AVC's and HEVC's block subdivision capabilities. Retrieved from Will HEVC reduce cost in the video business? (Bruno Zagorščak, 2016)



1.8 AV1

AOMedia Video 1 (AV1) was developed in 2018 by the Alliance for Open Media which is a non-profit industry consortium, focusing on the development of open, royalty-free multimedia technology, namely, open alternatives to standards developed by the Moving Picture Experts Group (MPEG). Currently, several industry leaders/big tech companies with significant market shares in their domains, are part of AOMedia. The governing members are Amazon, Apple, ARM, Cisco, Facebook, Google, Huawei, Intel, Microsoft, Mozilla, Netflix, Nvidia, Samsung Electronics and Tencent. [7][70]

The AV1 standard provides 3 profiles, Main, High and Professional, differing in available subsampling types, as well as color depths. Usage of different profiles mainly regulates video quality, dictating average bitrates at different resolutions and framerates [31]. Just like other video coding standards, AV1 uses blocks and divisions upon them. Pictures are divided into 128×128 or 64×64 blocks (called superblocks) first, then can be divided recursively as is needed. In contrast with HEVC, those blocks can range from 4×4 samples up to 128×128 . Different block sizes support different divisions, as can be seen in figure 6. Blocks containing 4×4 samples cannot be divided further. [70]

Figure 6: Block partitioning capabilities of blocks of different sizes in AV1. Note: diagram created by author using data retrieved from The AV1 Video Codec (Timothy B. Terriberry, 2019)



Most common desktop browsers, such as Google Chrome, Mozilla Firefox, Opera and Microsoft edge currently support AV1 format decoding, as well

as the VLC player on desktops, and the Android and iOS mobile operating systems. FFMpeg, a collection of open-source software for working with audio and video data, supports the libaom codec for AV1, and includes the libavcodec library.[30]

1.9 VVC/H.266

The newest standard as of 2024 is Versatile Video Coding (VVC). Finished and published in 2020, this standard aims to be versatile in regard to video use cases that it supports, as well as to once again provide further improvements to video compression rates.

VVC improves upon HEVC in several fields. Examples that can be mentioned include intra-picture referencing, where this standard doubles the amount of angles that the motion vectors can utilize in prediction, virtual boundaries, that in cases where they're used, specify where loop filters should and should not process the image, or more flexible block subdivision, where VVC introduces new ways in which pictures can be recursively cut into blocks. Section 3.3 goes into more detail on this topic.

2 Joint Video Experts Team, development

The **Joint Video Experts team** (JVET) is a team of specialists in video coding technology, formed by the Video coding experts group - a working group of ITU-T study group 16, and MPEG - a working group of ISO/IEC JTC 1/SC 29.[50]

The ITU-T Study Group 16 was formed in 1997, and is the department responsible for studies regarding various multimedia applications and systems, in the ITU-T - the department of telecommunication standardization in the International Telecommunication Union.[48][47]

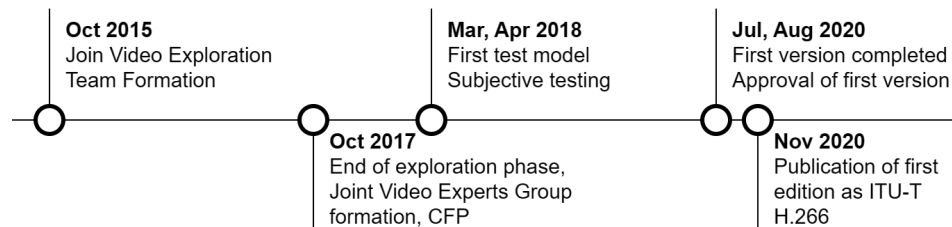
Formed in 1991, ISO/IEC JTC 1/SC 29 is a working group in the International Organization for Standardization (ISO), that focuses on creating standards in the field of efficient coding of digital information, visual or otherwise, as well as, among others, media security and privacy.[43]

2.1 History, development of VVC

After the High Efficiency Video Coding (HEVC) standard was finalized in June 2013, the authors continued to do research in the field, exploring new ways to develop the technology. After seeing some potential improvements with regard to HEVC, experts from the ITU-T Study Group 16, along with their colleagues from the ISO/IEC JTC 1/SC working group joined forces to explore new possibilities in the technology, formally creating the Joint Video Exploration Team in october of 2015. The purpose of this union was to involve the top experts on this topic in a synergistic effort, as well as to avoid duplicate activities in both organizations. [17][51]

During exploration, both the possibility to extend the most recent standard - HEVC, and the possibility to create an entirely new standard, were considered. [51]

Figure 7: A timeline of significant events regarding JVET. Note: diagram created by author using data retrieved from JVET - Joint Video Experts Team (ITU-T, 2020)



In march of 2017, the "Joint Call for Evidence on Video Compression with Capability beyond HEVC" was issued. [87] A call for evidence (CFE) is an information-gathering exercise that seeks expertise from people, institutions and other stakeholders with knowledge of a particular issue. [23] This resulted in a list of specifications for the new standard, with the initial designation of H.FVC - a "Future Video Coding" project. This list included a lot of features we see in VVC as a result, including, but not limited to, end-to-end delay optimizations, better compression or support for screen content. The Joint Video Experts team, keeping the same abbreviation, was formed following the approval of the CFE, in october of 2017. [50][49]

In the same month, "Joint Call for Proposals on Video Compression with Capability beyond HEVC" was issued. Calls for proposals are open invitations for funding of a project. [28] In the document, planned research was outlined in more detail. [71]

Next up, over March and April of 2018, subjective tests were performed on the test model that was developed so far. A total of 7 test labs, in italy and the UK, were involved in the testing, each experienced in their respective fields. Tests were partially overlapping, so that verification about the validity and comparability of the tests would be possible. Each lab was tasked with testing a specific type of video, in which the test sequences were subdivided. Tests were performed on SDR video in HD and UHD resolution, 360° video and HDR video in both PQ (Perceptual Quantization) and HLG (Hybrid Log Gamma) formats. The following labs participated in the testing:[9]

- BBC (England), Queen Mary University of London (England), University of West Scotland (Scotland) - SDR at HD resolution
- Centrum Wiskunde & Informatica (Netherlands) - SDR at UHD resolution
- University of Padova, GBTech (Italy) - 360° Video
- RAI, Sisvel (Italy) - HDR PQ
- DBW, EVATech (Italy) - HDR HLG

The number of human viewing subjects was kept high, in order to minimize fatigue, and keep the efficiency of responses at a good level. In total, over 2000 people participated in testing. 58 test session were performed, with work done exceeding 500 hours, including both laboratory operations and the tests themselves. [9]

The design of all the 58 test sessions was entirely done by the test coordinator, with the support of a CWI expert. The SDR testing effort required 36 test sessions in total, HDR tests were done in 14 test sessions, while 360 required a low 8 test sessions. The medium duration of a test session never went past 20 minutes; using this solution allowed a low fatigue level and high efficiency of the viewing participants. All scores were collected by paper sheets. A novel data acquisition technique was utilized (already successfully tested in other JVET testing experiments), consisting of assigning the data entry task to the viewers. This was both very safe and efficient (two viewers dialled in the data, other two checked the inputs) and allowed to lower the level of stress of the participants changing the cognitive load of the viewing activities.[9]

In summer of 2020, the first edition of the new standard, now named Versatile Video Coding (VVC) was finalized, and approved on the 29th of August thereafter. [16] Publication of this edition came later in November of 2020, as ITU-T H.266. A 500-page document, which includes the standard’s technological specifications, was published.

2.2 Industry adoption

The biggest players standing behind VVC in industry are members of the Media coding industry forum (MC-IF), whose mission has been to put forward and promote technology developed by MPEG for years. The list of significant members of this organization includes but isn’t limited to: Intel, Nokia, Dolby, Tencent, Orange, Ericsson and more. [63][22]

The organization expects VVC to appear in a growing number of mobile devices and other consumer electronics, such as smart TVs. Justin Ridge, the president of MC-IF has been quoted as saying *”Beyond publishing guidelines, 2023 was also a watershed year because several standards bodies approved VVC – including SBTVD, DVB, SCTE and ATSC (where it is in-progress). These approvals are important because it lays the foundation for practical deployments, as a result, we expect VVC to appear in a growing number of mobile devices and consumer electronics – especially smart TVs – as early as 2024. After that, we anticipate it to permeate into other areas, such as surveillance and security, in devices like drones and security cameras,”*, essentially marking 2024 as the breakout year for VVC. [22]

So far, according to a report by Bitmovin, VVC has enveloped a rather small bit of the market, standing at 7% share of live and VOD (Video on demand) encoding, the most significant rivals still being h.264 (AVC) and h.265 (HEVC), along with AV1 sharing around 7% of this market as well,

however, there has been a recorded uptick in developers planning to use AV1 for VOD and streaming, more so than VVC. [22]

In their report, Bitmovin also stated that "Adoption of newer generation codecs is still disappointingly low, but AV1 seems to have the most optimism around it for the coming year, likely driven by Apple adding support to its latest mobile and desktop processors." [22]

3 Technology Overview

The purpose of this section is to explain concepts relevant to VVC's technological capabilities, as well as to explain what improvements it has made over the preceding, HEVC video coding standard.

As the name suggests, VVC was designed to be truly versatile. This means considering the current and future developments in the needs of users towards video coding technologies, to which will be referred to in this section. Among the considerations of the researchers was also to re-explore already used video coding techniques, and make improvements to them, as users all around the world continue to increase their usage of this technology, and benefit greatly from upgrades to it.

3.1 Alternative types of video streams

The team behind VVC was mainly focusing on adapting to media trends that are significant or could be in the future, orienting to the future of the field. [15] Trends, such as spherical video (used in conjunction with virtual reality headsets), the popularity of livestreaming, especially associated with video gaming and its video streams, or the popularity of video with quite significant increases in color depth (like HDR). This section will give brief descriptions of these alternative forms of video, which some of VVC's features are tailored to support.

3.1.1 360° Video

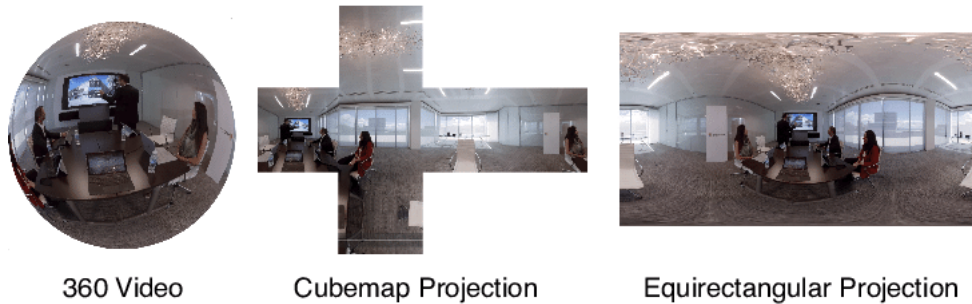
Online streaming of Virtual Reality and 360° videos is rapidly growing, as more and more major content providers and news outlets adopt the format to enrich the user experience. - [6]

360° video, also known as immersive video, spherical video or Cine-VR[27], is a type of video format that extends its content in a way to allow the user to look around freely while watching - it includes all the information to envelop the user fully in its contents. This type of video is usually filmed on either an omnidirectional camera, or a collection of cameras whose recording is spliced together. Typically, this content can be viewed on a device that is capable of displaying traditional, rectangular video, by allowing users to pan around and change their viewing angle. However, where this type of video truly shines, is when viewed through a virtual reality headset, enveloping both the users eyes and ears. Using this technology, the user's head movements are tracked, and directly translated into movement within the 360 video, allowing them to feel truly immersed in the content, sometimes even causing feelings of fear, dizziness or vertigo. [60]

360° video has been getting popular since around 2015. In that year, Facebook started to support the playback of this type of video, along with YouTube, which launched this ability in the same year. In 2016, users were enabled to upload their own 360° content on the website as well. Since then, a number of news outlets, such as, among others, the BBC or The New York Times, started occasionally using this format for select coverage.[72]

There are several ways in which 360° video is stored - it all boils down to representing the video data necessary to form the, at a first glance, three-dimensional video format as a two-dimensional video stream that is friendly to traditional video coding techniques. In figure 8, we can see two kinds of representations of immersive video on the same sequence, one being the Equirectangular projection - a projection that is similar to how various geographical maps are commonly represented on two-dimensional media, the other being the cubemap projection - which treats the saved media as walls of a cube being looked at from the inside.

Figure 8: A showcase of different spherical video projections. Retrieved from Towards Generating Ambisonics Using Audio-Visual Cue for Virtual Reality (Rana, Aakanksha and Ozcinar, Cagri and Smolic, Aljosa, 2019)



3.1.2 High Dynamic Range video

High Dynamic range (HDR) media, is media that works with a wider range of colors than typical media, referred to as Standard dynamic range (SDR). Increasing this range means providing more resolution to colors used in the footage, helping portray the subject of the media in a wider degree of colors and brightness levels. SDR first came at a time when technology was limited in its ability to display a wider selection of colors and brightness levels, an example would be CRT monitors. However, this range is still widely in use even in modern devices.[19]

HDR video requires the medium on which it's being displayed to support this technology, and along with it, it creates a more immersive viewing

experience. The advantages of this format include: [19]

- the ability to reproduce more detail in areas of the image that are overall significantly lighter or darker than the rest of the image,
- the ability to create less jagged, and more convincing and smooth gradients between colors,
- the ability to more accurately represent the subject of the video, having more colors to choose from.

The technical character of this type of video data is usually done quite simply by increasing the size of the memory allocated for the data types used to store the values of the colors in the image - therefore making this video format more memory-heavy. Typical SDR applications use 8 bits per color channel, resulting in the usage of 24 bits per pixel, with a total of over 16 million colors available. HDR, however, typically uses 10 or 12 bits per channel, significantly increasing the total number of colors to over a billion.[19]

HDR can also implement transfer functions, such as Perceptual Quantization (PQ) or Hybrid Log Gamma (HLG). The purpose of these functions, developed by Dolby Vision and BBC, respectively, is to better match the human visual system, by attempting to use the available color resolution more efficiently.[68]

Perceptual quantization aims to be the most precisely tailored way to get light out of an electronic signal. A single signal level change should, by this standard, cause a step in brightness that is just below the threshold of being visible, at all points between the darkest black and the brightest white (at a certain color resolution and display brightness capability).[68][12]

HLG does things differently. Its gamma values aren't based on display brightness capability, but scene brightness (brightness derived from video footage alone), meaning signal content is constant with changing available displays. The biggest difference between HDR with HLG and traditional SDR signals is in brighter areas of footage. HLG reserves a lot more resolution for signals with higher brightness, while acting very similar to SDR on lower brightness ones.[68][12]

3.1.3 Wide color gamut

Wide color gamut (WCG) specifically aims to increase the available range of colors that a viewing device can display, in order to portray reality in a more life-like way. Along with HDR, both technologies have been tested to

impact viewing experiences more than just increasing the resolution from 2K to 4K. Implementations of both technologies are often found together on newer viewing devices like TVs, computer monitors, and more.[65]

3.1.4 Computer screen content

As the digital content space evolves, whether for educational, entertainment, or other purposes, more and more content on the internet comes from recording one's screen. This content often features computer generated graphics, text etc. Among others, applications include video game streaming or online education. [88] In fact, approximately 1.86 billion hours of video game streaming content was being watched (as of 2022) by users every month on Twitch, a popular video game streaming service. [25]

The design of conventional video codecs typically caters to traditional camera-captured video footage, but that exhibits significant differences from screen content, in its signal characteristics, and what areas need to be supported to provide humans the best quality experience.[88]

3.1.5 Ultralow-delay applications

This describes applications that favor as minimal a delay in video transmission as possible, which is convenient in several very popular use cases of this technology. As I'm sure we're all aware, video conferencing has seen a huge increase in use as of the Covid-19 pandemic. In fact, the use of video conferencing applications grew by 62% during the pandemic.[56] As is obvious, users prefer to experience as little latency as possible while conferencing. Another use case that benefits from ultralow-delay optimizations is livestreaming, as referred to above. Most livestreaming platforms offer a way for users to interact with the host, sending messages in chat, or via donations. Of course, those users prefer to have as little delay as possible in these channels as well.

What happens in these scenarios is that the video gets encoded, sent and decoded. Therefore, it is especially favorable here to allow coding and decoding to be faster, and for the video data sent to be as small in size as possible. However, recent advances in network technologies, such as WiFi-6 or 5G, lowering the latencies to milliseconds, the video coding pipeline becomes the main bottleneck [84]. Stability of the video on the decoding and displaying end is also highly important, so sometimes it's beneficial to temporarily sacrifice picture quality or video frame rate to achieve it.

3.1.6 Beyond HD video

This refers to video of extraordinary high resolution, specifically, for example, 4K and 8K video. Steady advances in both display and video recording technologies make these formats more and more common in everyday applications. Capture of 4K video is becoming common even on mobile devices such as smartphones. These types of video streams disproportionately increase the load on data transmission and storage, so they especially benefit from better compression technology.[15]

3.2 Common video coding concepts

In the field of video coding, you can find features, that almost all video coding technologies use to compress video. When it comes to video streams, in many ways, a lot of these features are similar in character to R.D. Kell's 1929 patented idea, alluding to focusing on the changes between frames as opposed to all of the displayed information in pictures. More techniques, that focus on saving space within singular pictures, are also used.

This subsection will focus on some of these features/concepts that are more relevant to improvements that VVC has made over the previously latest technology.

3.2.1 Inter-picture referencing

Inter-picture referencing refers to a compression technique that stores frames, called predicted frames, as information about differences from another, called reference frames. This is partly done with the use of a technique called Motion Vectors, which carry information about two-dimensional translations within pictures. Essentially, predicted frames are compared against their reference frames using a block-matching algorithm - an algorithm that tries to find similar areas in the two pictures, and establish that this area simply moved elsewhere between the frames. How much and in what direction it moved is stored as motion vector information. What other differences between the areas still are, called prediction error, can be calculated and transmitted separately. [32][75][13]

As the motion between two pictures can be arbitrary, motion vectors benefit from being as flexible as possible, to represent the source material well, and most standards have a set number of defined angles at which the vectors can transform the blocks. For example, in figure 9, HEVC's available motion vector angles are on display. [78]

As mentioned, in inter-prediction, different types of pictures are used. Reference frames, also called I-frames (Intra-frames) are pictures that are stored fully, without any inter-frame prediction, therefore they can be decoded independently from other pictures. From these, predicted frames, or P-frames are constructed using motion vectors. As opposed to I-frames, they cannot be decoded independently, since they rely on the I-frame information. The third common picture type is a B-frame (Bi-predicted frames). Those use information from two other frames to take shape, using both I-frames and P-frames as references. Figure 10 shows how these frames are commonly used in video coding. After each I-frame, a set of pictures that reference it in prediction are present. Those pictures, along with the I-frame are called a Group of Pictures (GOP). The longer the GOP is, the fewer I-frames are present, saving memory. However, each I-frame refreshes the stream, so to speak, so a longer period without one will cause the stream to devolve into unintelligibility. [32][13][62]

Figure 9: HEVC Motion Vectors. Retrieved from *Fast Intra Mode Selection in HEVC Using Statistical Model* (J. Tariq and A. Alfalou and A. Ijaz and H. Ali and I. Ashraf and H. Rahman and A. Armghan and I. Mashood and S. Rehman, 2021)

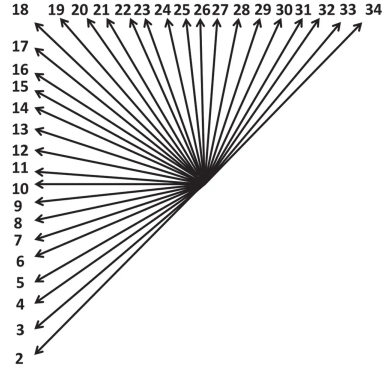
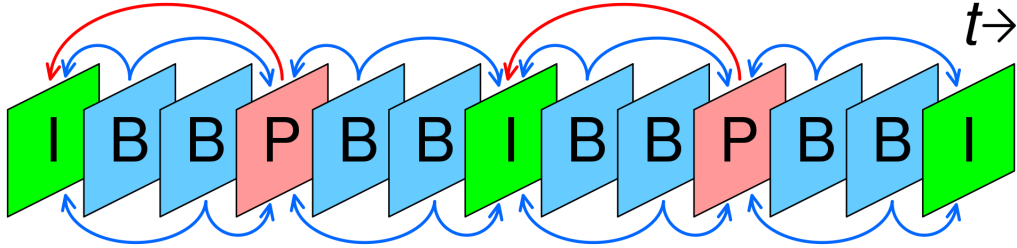


Figure 10: How predicted frames are created from references, based on type. Retrieved from *Illustration of dependencies of the group of pictures scheme IBBPBB...* Time goes from left to right. (Cmglee, 2018)



3.2.2 Block partitioning

Almost all video coding standards feature block partitioning in some way. Before HEVC, macroblocks have been used as a basic compression unit for video coding, with little capability for further partitioning, if any. As popular video resolutions reached full HD, and kept increasing, the macroblock-based design wasn't holding up in efficiency, and trying to improve on it wasn't yielding satisfactory results. For this reason, HEVC has later done

away with this design, in favor of its new recursive block partitioning structure, which provided the most significant coding gains over AVC. [42] Differences between AVC and HEVC's block partitioning can be seen in Figure 5.

VVC inherits this technology, on which it improves in ways elaborated on in section 3.3.

3.2.3 Loop filters

Block-based video coding is known to suffer from block compression artifacts, primarily on block edges. To attempt to fix this, deblocking filters are used in most block-based video coding standards. [1]

A different type of loop filter, which is used in HEVC, is the Sample adaptive offset (SAO) filter. The SAO is performed after the deblocking filter in the video coding loop. Conceptually, SAO classifies reconstructed pixels into different categories and then reduces the distortion by simply adding an offset for each category of pixels. The pixels are categorized using intensity and edge properties. To further improve the coding efficiency, a frame can be divided into regions for localization of offset parameters. [33]

Another type is the "film grain synthesis" filter, used by AV1. Film grain in video is naturally opposed to efficient video compression, due to the high-density detail on a low scale that it causes, and therefore is hard to code with a measure of quality, and can actually cause worse compression artifacts. AV1 addresses this with this filter, which works by analyzing the specifics of the film grain, storing their parameters, and before encoding, completely removes this component from the video. The decoder then receives this information along with the video, and adds it back in afterwards. [66]

3.3 Improvements upon HEVC

This section will go over some of the main feature improvements upon the last video coding standard by the organizations involved, HEVC of 2013. Versatile video coding tries to improve in areas deemed insufficient in HEVC, and as alluded to, tries to cover a wider spectrum of video technology use cases in industry.

3.3.1 Random access capability

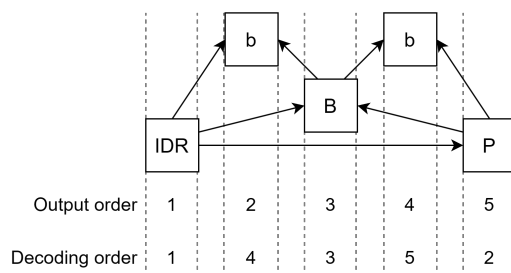
Random access capability is the ability to start watching a piece of video footage at an arbitrary timestamp (e.g. seek to 3 minutes into a YouTube video, change the program on a TV, or change between bitstreams based on

network connection quality), and not just the beginning of the stream. It supports very basic video viewing features, that most applications require to function. [15]

Those not new to the field of video coding can probably already sense the problem. It's rare today to find in use a video codec that doesn't use inter-frame coding to encode video footage. Usage of this technology means that frames of video depend on one or more other frames with their visual data. This is where two different orderings of frames come into play - the output order, and the decoding order. The output order simply refers to the order in which frames of video are displayed to the user - it is the original order of frames in which the video footage has been created. The decoding order puts frames in sequences based on which frames' information is needed to create frames. For example, if frame 3 (in a numbered sequence in ascending order) is a P-frame (see paragraph Inter-picture referencing) referring to 1's data, 1 has to be decoded first in order for 3 to be created. If 2 is a B-frame of 1 and 3, both frames 1 and 3 have to be decoded before 2 can be created. This means that frame 1 has to precede frame 3 in the decoding order, and both frames 1 and 3 have to precede frame 2 in the decoding order, resulting in a decoding order of 1,3,2, despite the output order being 1,2,3. Figure 11 shows how the output order might differ from the decoding order on a small example. [15]

For random access, the simplest feature VVC uses is the instantaneous decoding refresh (IDR) picture. An IDR picture is characteristic by not depending on any other pictures, and thus has the ability to be decoded alongside and all pictures depending on it without any information from preceding frames. IDR pictures, however, are by definition a big cost in coding efficiency by interrupting inter-frame coding. A way to perform random access more efficiently is the clean random access (CRA) picture. The difference between it and the IDR picture is that there are pictures that depend on the CRA, that follow it in decoding order, but precede it in the output order, therefore the decoding of these images has to be skipped, since they would have been shown before the CRA

Figure 11: Decoding and output orders. Note: diagram edited by author from a diagram retrieved from Overview of the high efficiency video coding standard (Sullivan, G.J. and Ohm, J.R. and Han, Woo-Jin and Wiegand, T., 2021)



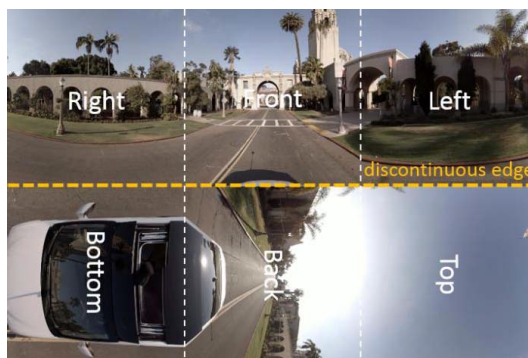
picture if the stream was played normally without random access. The efficiency savings come from the CRA pictures not interrupting inter-frame coding. Both of those types of pictures are referred to as intra random access points, or IRAP pictures. [15]

This, however, still bears the cost of coding those pictures in full fidelity, without using inter-picture prediction (since that is why they are used for random access). That comes with a memory cost, that both affects the total file size of the footage (and therefore network condition requirements), and the decoding speed. So another tool at VVC's disposal is the gradual decoding refresh, or GDR, picture. As arguably the most complex, but most efficient way to do random access, the GDR picture can depend on other pictures through inter-frame prediction, unlike an IRAP picture. This fact may cause the GDR picture not to be completely correct in its data - due to this incompleteness of information. For this reason, the stream might decode to incorrect information, up until a recovery point picture, that doesn't depend on previous pictures in its content, is reached. Typically, a refreshed region starts at the GDR picture, and grows gradually to include the entirety of the frame, by the time the recovery point is reached. [15]

3.3.2 Virtual boundary

Refers to technology that modulates the use of in-loop filters, that usually work through the entire picture. Its purpose is to limit their effect across specified paths, mostly in situations where video footage is being interpreted differently than usual. For example, in the case of a cubemap projection (see section 3.1.1), faces that are to be wrapped around a virtual cube are portioned into 2 sections, as seen in Figure 12. Filtering on the boundary would create unwanted visual artifacts. This usage take advantage of this technology's static variant, but a dynamic one is available as well, which is used with GDR pictures. [15]

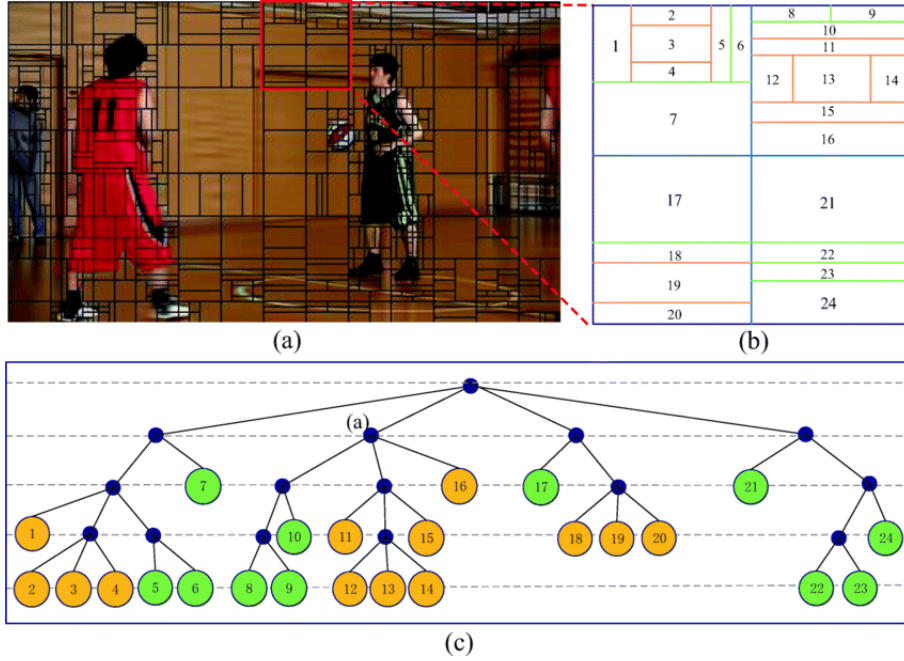
Figure 12: Discontinuous edge in 360° video. Source: [15]DESCRIPTION. Retrieved from Overview of the Versatile Video Coding (VVC) Standard and its Applications (Bross, Benjamin and Wang, Ye-Kui and Ye, Yan and Liu, Shan and Chen, Jianle and Sullivan, Gary J. and Ohm, Jens-Rainer, 2021)



3.3.3 Improvements to the flexibility of block partitioning

VVC makes several improvements to the flexibility of the block partitioning system. The largest blocks the pictures can be split using were increased in size from 64×64 to 128×128 samples. More splits were also introduced, being binary and ternary splits, each with horizontal and vertical variations. This allows for a better adaptation to image content for high spatial resolutions while also supporting small and narrow rectangular blocks for the prediction and transform processing stages. [17] Figure 13 shows some examples of how a picture could be split and represented in trees, using these new capabilities.

Figure 13: Example of block partitioning in VVC. Retrieved from *A fast CU size decision algorithm for VVC intra prediction based on support vector machine* (Chen, Fen and Ren, Yan and Peng, Zongju and Jiang, Gangyi and Cui, Xin, 2020)

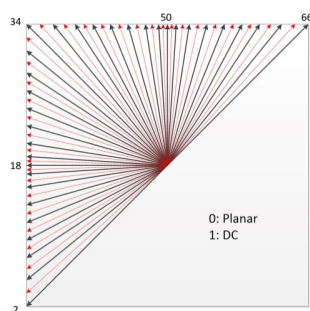


3.3.4 Improvements to motion vector capability in intra-prediction

Motion vectors in intra-frame prediction have received an upgrade in terms of flexibility. The angles this technique can use in this standard has been roughly doubled, from 33 to 65. [81] Figure 14 shows the possible angles, in comparison to HEVC.

3.3.5 Scalability and Layered Coding

Figure 14: Intra-prediction angle improvements. Retrieved from VVC codec review, comparison with HEVC (Dmitriy Teplyakov, 2021)

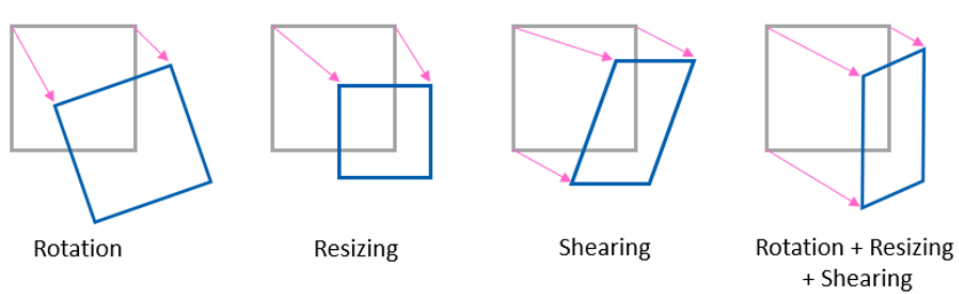


Scalable video coding refers to defining the structure of a coded video stream so that a subset of the data can be extracted, for example containing the same content but at lower visual or temporal quality, or including supplemental decoded content. Temporal quality mainly refers to video framerate and continuity. For example, this can be used to momentarily lower video framerate to deal with unstable network conditions, or conditions under which different clients are differentiated by their connecting bandwidths - subsets used for this would be referred to as temporal sublayers. [15]

3.3.6 Affine Motion

In intra-frame prediction - a technique that uses pixels from the current frame to construct other pixels in the same frame, VVC introduces Affine Motion, a feature similar to one in AV1, and a newly introduced one in regard to HEVC. Most previous coding standards only use two-dimensional translations along the plane, something referred to as 2 degrees of freedom (DOF). VVC introduces 4 more total DOF, in rotation, scaling, aspect ratio changes and shearing. In figure 15, examples of those transformations are displayed. [83]

Figure 15: Affine motion transformations. Retrieved from *Affine Motion Compensated Prediction in VVC* (Vicuesoft, 2021)



4 Methods of objective video quality comparison

This section goes over methods of measuring and comparing objective video quality, which will be used in the next section to do so with regard to VVC.

Video quality can be judged both objectively and subjectively. Subjective testing usually consists of researchers putting together random groups of people, and organizing viewing sessions, in which participants are showed different pieces of video footage, and are asked to fill out questionnaires about what they think about the video quality. Such testing was also performed on VVC during development in 2018 (2.1). Objective metrics, on the other hand, are computed and measured by computers.[54]

Objective video quality models are used to predict results from subjective quality research - tests performed on live human audiences that rate the quality of video footage. They are mathematical models, and the common grouping they fall into is based on how much reference information they require, the categories are such:

- Full Reference (FR) - The model has access to a raw (non-compressed), high-quality version of the sequence, that it compares to the compressed version,
- Reduced Reference (RR) - The model doesn't have the raw reference, and does the best it can by computing key quality parameters from both sequences, and comparing,
- No Reference (NR) - The model doesn't access the reference sequence, and usually only judges aspects like compression artifacts and the likes.

[54]

4.1 VMAF

VMAF, or Video Multimethod Assessment Fusion, is a method for use to predict the results of subjective quality testing of a video stream, developed jointly by Netflix and a research group from the University of Southern California. As the name suggests, this method combines several elementary quality metrics, and assigns them weights in computing the final score. For this, a machine-learning algorithm was used, specifically a Support Vector Machine regressor. Training and testing of the model was based on a dataset of scores gathered through experiments with subjective viewing (the NFLX Video Dataset). The version of VMAF numbered 0.3.1 combines two elementary metrics, Visual information fidelity (VIF), and Detail loss metric (DLM). [11]

VIF is commonly used in the field to measure video quality. Originally developed at the University of Texas, it is now deployed as a part of VMAF, slightly modified. Without going into too much detail, this metric performs visual fidelity differences between the reference and distorted picture, and does so on multiple scales. [11]

DLM is featured in VMAF with slight changes as well, such as the handling of special cases, like black frames, where the original formulas yield unusable results. DLM separates the measure of detail loss that affects the visibility of the content, and the measure of distracting visual impairment in the footage. [11]

To also take into account temporal, or time-based characteristics, VMAF introduces another metric that focuses on motion. The goal is to simply calculate the temporal difference between subsequent frames, and is only performed on the luminance component of the video. [11]

4.2 MS-SSIM

Multi-Scale Structural Similarity (MS-SSIM) is based on the rationale that the human visual system is highly adapted to spotting patterns, to extracting structural information from visual stimuli. MS-SSIM tries to come to a measure of structural similarity between two images, and use it to provide a good approximation of perceived image quality in subjective testing. Separating itself from similar methods used previously, which usually compare images at one scale, MS-SSIM takes multiple scales into account, citing different conditions under which a person could view the medium, mainly from what distance the viewing happens. [90]

Figure 16: Multi-scale structural similarity measurement system. Retrieved from *Multi-scale Structural Similarity for Image Quality Assessment* (Zhou Wang, Eero P. Simoncelli and Alan C. Bovik, 2003)

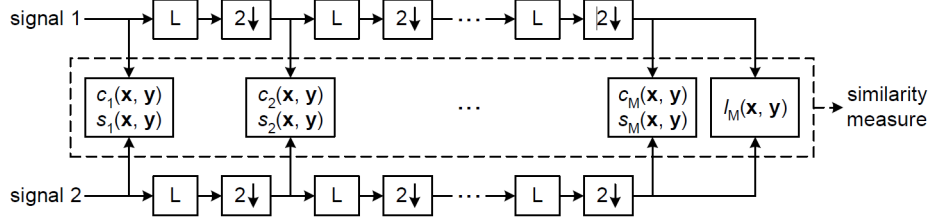


Figure 16 shows the method ms-ssim uses. It essentially performs tests on sets of images generated as downsamplings of the reference and the distorted version, taking all measurements performed into account when calculating the final similarity value.

4.3 PSNR-HVS-M

Peak signal-to-noise ratio (PSNR) describes the ratio between the signal and noise - signal being the information being transmitted, noise being an impeding factor on the information quality. Thus, higher values indicate higher information quality, or, in the context of visual media, better image quality. [24] In multimedia, PSNR is often used to measure image quality loss with the use of lossy compression, and its values measured on a logarithmic scale using decibel units.

PSNR is essentially an extension of a mean square error (MSE) calculation. To calculate the MSE between two images, the following formula is used:

$$MSE = \frac{1}{m \cdot n} \sum_{i=0}^{m-1} \sum_{j=0}^{n-1} [I(i, j) - K(i, j)]^2$$

Then, the final PSNR value is calculated with:

$$PSNR = 20 \cdot \log_{10}(R) - 10 \cdot \log_{10}(MSE)$$

This operation puts the value on the logarithmic scale, and adjusts the value based on the maximum signal range, represented by R . For SDR, for example, where color channels are represented by bytes of data, R would be 255.

[61]

PSNR has been built upon several times to produce a better metric. PSNR-HVS is such a technique, which tries to adapt better to the human visual system (HVS), and thus produce testing results that more accurately approximate video quality results that would be collected from subjective quality testing. For this, it uses a modified version of the Universal Quality Index (UQI). [26]

PSNR-HVS-M is a technique, that improves PSNR-HVS by taking into account an effect called visual masking. [26] This effect originates in the quantization function when DCT is used for compression. [53]

5 Objective comparison of quality

All data measured by author, that was used to construct charts in this section, is included in Appendix 1.

5.1 General parameters

5.1.1 Hardware, software

All encoding and decoding was performed on a system with manjaro linux OS, with specifications described in below:

OS Manjaro 23.0 – xfce version

CPU intel core i5-9400 2.60GHz

GPU MSI GeForce GT 710

RAM 16 GB DDR4 @ 2666 MHz

VVenC version v1.11.1

VVdeC version v2.2.0

ffmpeg version v7.0

vqmt v1.1

Ffmpeg, an open-source software project consisting of a suite of libraries and programs for processing a wide array of multimedia data [29], was used to perform all encoding and decoding, as well as testing quality with the VMAF method. As ffmpeg doesn't yet natively include tools for working with Versatile video coding, an installation integrating it with the vvenc encoder and vvdec decoder had to be gone through with [55].

5.1.2 Video sequences

For the comparison, 3 video sequences were chosen, which correspond to sequences in [41], with the purpose of comparing quality testing results to data from other previous coding technologies. The video sequences "Crowd Run" and "Park Joy" are available at [3], the sequence "Sintel" is available at [2].

Name	Resolution	Bit depth	Frame rate	Frame count
Crowd run	3840×2160	8-bit	50	500
Sintel	4096×1744	8-bit	24	240
Park Joy	3840×2160	8-bit	50	500

As the Sintel sequence was an entire short film, 10 seconds starting from 04:07.000 were used, to correspond to mentioned data. All sequences used are 10 seconds long.

5.1.3 Bitrate, presets

Bitrates at which to encode were chosen so that they correspond to data from [41]. According to the work, starting at 1000 kbit s^{-1} was too low, and some codecs weren't capable of going this low, instead opting to use around 2000 kbit/s at the least. Ending at $20\,000 \text{ kbit s}^{-1}$ was too low as well, as the codecs tested didn't seem to reach good enough results.

In the end, 16 different bitrate values were chosen.

- Between 2000 kbit s^{-1} - $10\,000 \text{ kbit s}^{-1}$, with an interval of 1000 kbit s^{-1}
- Between $12\,000 \text{ kbit s}^{-1}$ - $24\,000 \text{ kbit s}^{-1}$, with an interval of 2000 kbit s^{-1}

Thus, the final recounting of bitrate values goes as follows: 2000, 3000, 4000, 5000, 6000, 7000, 8000, 9000, 10000, 12000, 14000, 16000, 18000, 20000, 22000, 24000.

As for presets, the installed vvc encoder has 5:

- faster
- fast
- medium
- slow
- slower

For all encoding of the "Crowd Run" sequence, the "slower" preset was used, as it should provide the most quality video with similar compression rates. For the other sequences, "Sintel" and "Park Joy", the second slowest "slow" preset was used, as it substantially cuts coding time while losing almost no quality to the "slower" preset.

Additionally, The videos needed to be in raw formats for the comparison tools to work on them, so all videos were, after encoding, decoded to the original y4m format.

5.1.4 Commands used

The following commands have been used to encode, decode and test video sequences. These commands were then incorporated into scripts that, alongside running them, created logs of the process. VMAF values were read straight from the console, thus no output file.

VVC Encoding:

```
ffmpeg -benchmark -y -i $ORIGINAL.y4m -c:v vvc -preset slower -b:v $BITRATE -threads 6 -an $ENCODED.mkv
```

VVC Decoding:

```
ffmpeg -i $ENCODED.mkv -pix_fmt yuv420p $DECODED.y4m
```

VMAF:

```
ffmpeg -i $ORIGINAL.y4m -i $DECODED.y4m -lavfi libvmaf -f null -
```

VQMT:

```
vqmt $ORIGINAL.y4m $DECODED.y4m $HEIGHT $WIDTH $FRAME_COUNT 1 $VQMT_OUTPUT PSNRHVSM MSSSIM
```

Parameters:

\$ORIGINAL - the source video sequence as a raw video

\$ENCODED - a version of the source video, encoded using the VVenC encoder

\$DECODED - the encoded version decoded back to raw video to compare

\$BITRATE - the desired bitrate value

\$HEIGHT - pixel height - number of pixels along the height of the video sequence

\$WIDTH - pixel width - number of pixels along the width of the video sequence

\$FRAME_COUNT - the number of frames in the video sequence being compared

\$VQMT_OUTPUT - the prefix used by vqmt when creating output files, logging quality values for each frame, calculating the average at the end of the files.

5.1.5 Codecs used

The data which will be compared against, from [41], performs tests on several other codecs, which range in recency and complexity. From codecs that the work tests, that implement the av1 video coding standard, the most performant codec in terms of quality was chosen, which also measures data for all the video sequences, along with the SVT-AV1 video codec, as an implementation by different authors with results that are also good. Additionally, all the data from the other video coding standards being compared against is used in this thesis. The list of codecs data from which is used goes as:

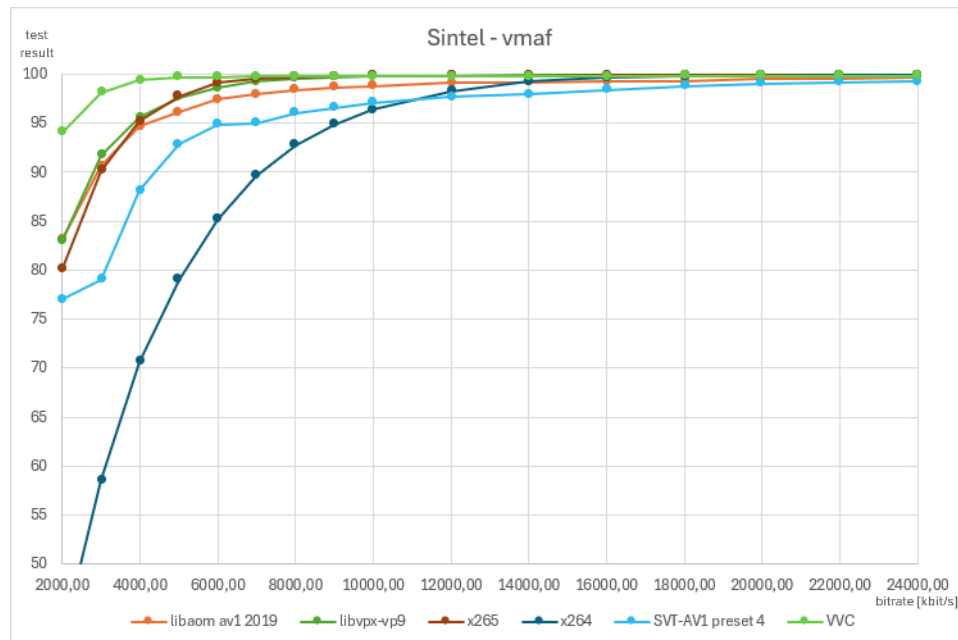
- Libaom AV1 2019 - the 2019 version of this codec, which maximized quality at the cost of substantial coding time
- SVT-AV1 preset 4 - the SVT-AV1 codec with the preset 4, which makes a compromise between coding speed and quality
- libvpx-vp9 - a codec that can be considered a predecessor to AV1, and, at the time, in direct competition with HEVC
- x264 - an implementation of the h.264 video coding standard, also known as AVC
- x265 - an implementation of the h.265 video coding standard, also known as HEVC

5.2 Quality comparison

5.2.1 VMAF - quality comparison

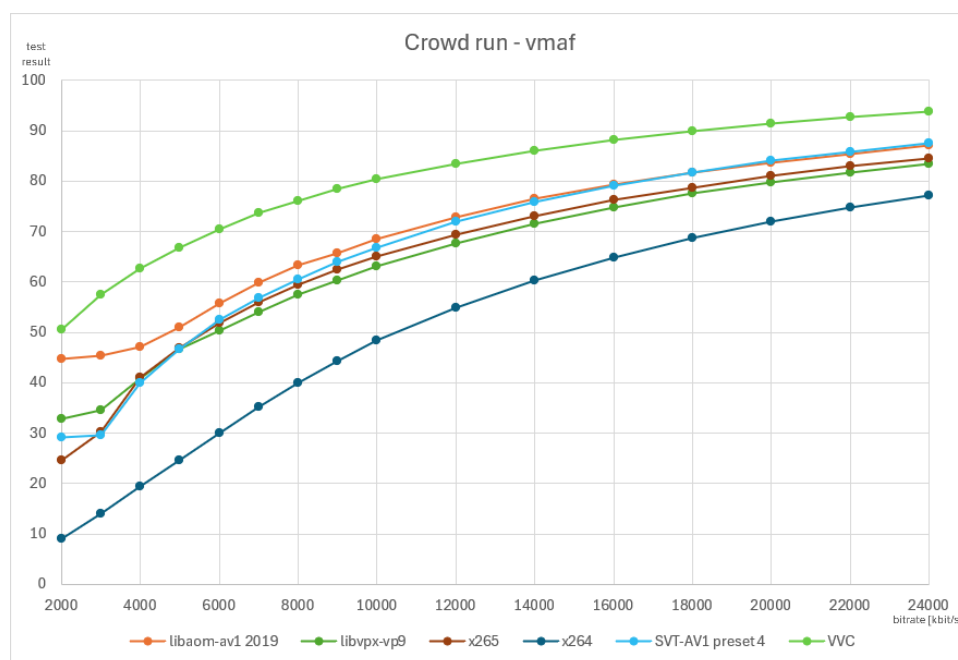
Following are the results of testing VVC with the VMAF method referred to in 4.1, performed on all 3 sequences. For reference, the range of VMAF scores is 0-100, more being better.

Figure 17: The results of the VMAF tests performed on the "Sintel" sequence. Note: chart created by author using measured data, along with data retrieved from *Objektivní porovnání kvality video kódovacího formátu AV1* (P. Hovorka, 2021)



From the shape of the curves in figure 17, we can see that all codecs struggle on very low bitrates, as would be expected, though some more than others. As the oldest codec, x264 quite significantly lags behind other codecs as the footage coded by them improves with higher bitrates. SVT-AV1 on its preset 4 shows the compromise between coding quality and time that it utilizes, as the quality of its resulting footage are second worst. Following are the libvpx-vp9, x265 and libaom-av1's 2019 version, all three in a close space. VVC seems to top the chart, as it's quality results are quite good at even significantly limited bitrates, and it achieves almost perfect results at a bitrate of just 4000 kbit s^{-1} . It's perhaps important to point out that the Sintel sequence is entirely computer-generated, altering the nature of the bitstream.

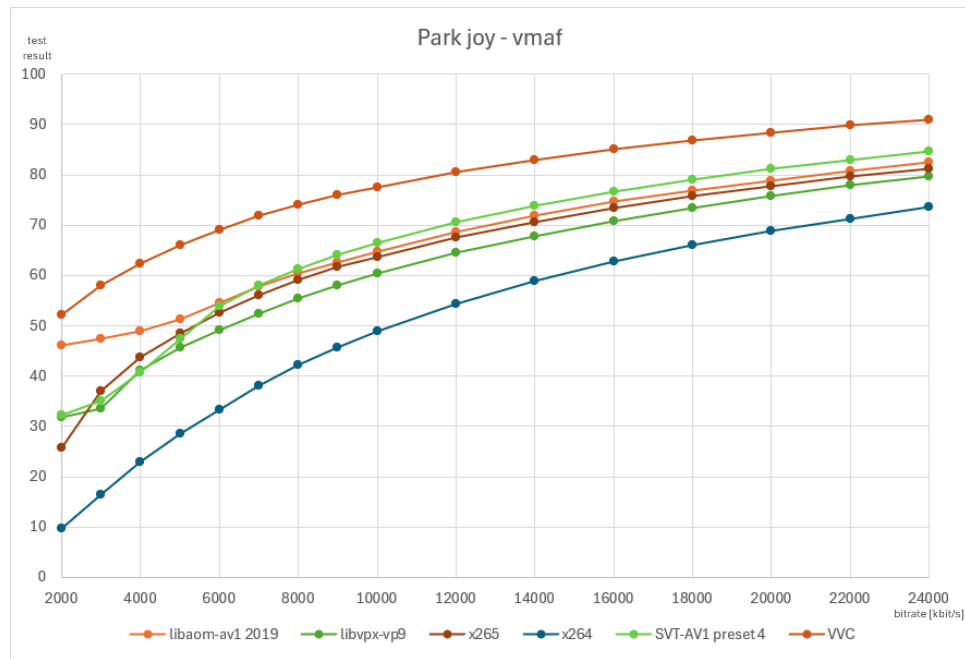
Figure 18: The results of the VMAF tests performed on the "Crowd Run" sequence. Note: chart created by author using measured data, along with data retrieved from *Objektivní porovnání kvality video kódovacího formátu AV1* (P. Hovorka, 2021)



Looking at figure 18, testing the "Crowd Run" piece of footage, we can see that the quality measurements act quite differently when encountering camera-filmed footage, such as this one. For example, the x264 codec returns almost incomprehensible video on the lowest bitrates. Other codecs do better, but still quite bad at the lower bitrates. Following x264 in the order of lowest to highest quality are x265, SVT-AV1 on preset 4, libvpx-vp9 and libaom-av1's 2019 version, which stand quite close to one another in their

quality improvements, as the allowed bitrate increases. VVC does quite well on this test, scoring above all the other codecs at all bitrates at a significant margin. At the full 24 000 kbit s⁻¹, it is nearing perfect scores on the VMAF quality scale.

Figure 19: The results of the VMAF tests performed on the "Park Joy" sequence. Note: chart created by author using measured data, along with data retrieved from *Objektivní porovnání kvality video kódovacího formátu AV1* (P. Hovorka, 2021)

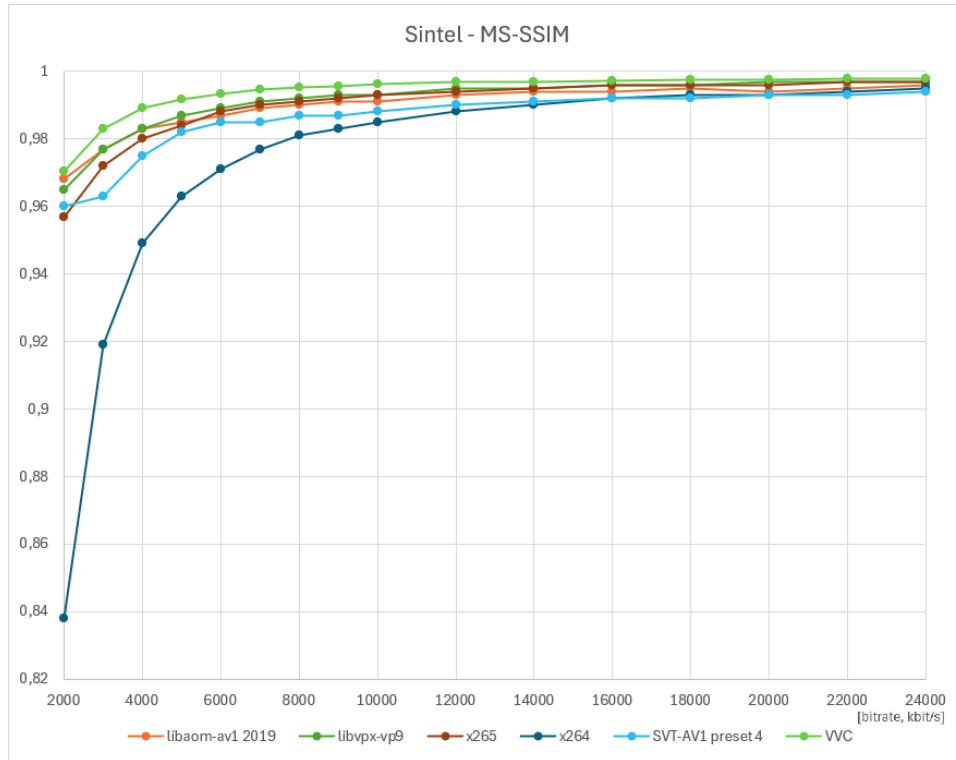


In figure 19, results of VMAF tests on the "Park Joy" sequence. Similarly to "Crowd Run", this is a camera-filmed piece of footage from the real world. Perhaps as a result, the curves look quite similar. Again, x264 is the lowest scoring, producing unusable video footage at the lowest bitrate values. x265, SVT-AV1 on preset 4, libvpx-vp9 and libaom-av1's 2019 version again score quite similarly to each other, more so as the bitrate values rise. Interestingly, libvpx-vp9 and SVT-AV1 produce a strange dip at lower bitrate values, being surpassed by x265 a few times. libaom-av1's 2019 version seems to also do significantly better at lower bitrates, in relation to other codecs.

5.2.2 MS-SSIM - quality comparison

Next up, results of the MS-SSIM tests on all three sequences. MS-SSIM results range from 0-1, more being better.

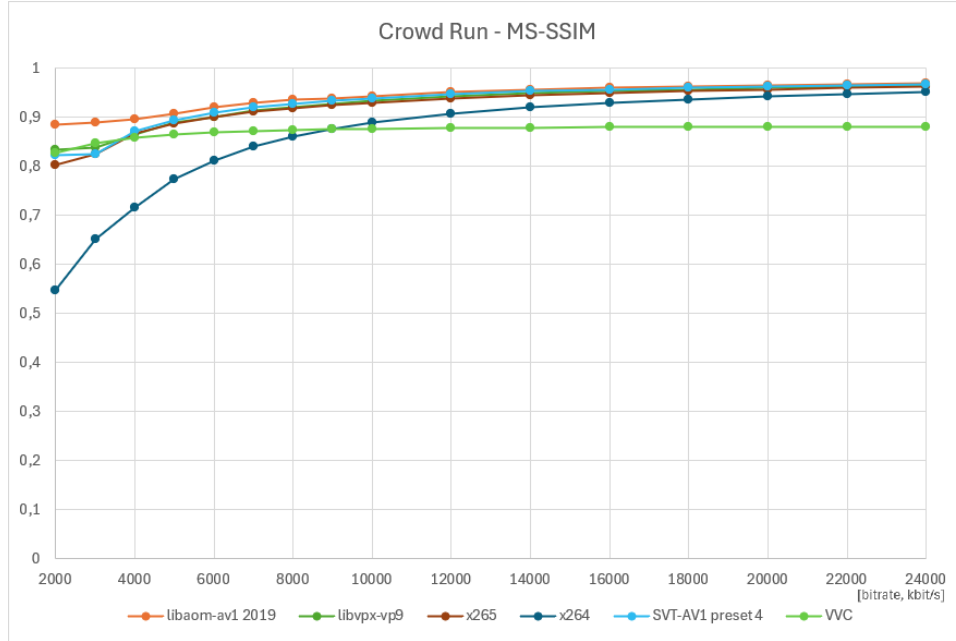
Figure 20: The results of the MS-SSIM tests performed on the "Sintel" sequence. Note: chart created by author using measured data, along with data retrieved from *Objektivní porovnání kvality video kódovacího formátu AV1* (P. Hovorka, 2021)



First, the "Sintel" sequence, figure 20. As we can expect, in the last place, x264 shows up again, with significantly lower scores, especially in the lower bitrates, not coming close to the newer codecs until reaching about $8\text{-}10\text{Mbit s}^{-1}$. The newer codecs can be seen following each other somewhat closely. With the second to worst score at 2000 kbit s^{-1} , x265 already has about a 0.96 score, which is quite good. SVT-AV1 on its preset 4 seems to exhibit an unexpected boosting of quality at 2000 kbit s^{-1} , leveling itself with the other codecs as bitrate values rise. VVC again comes out on top, achieving higher qualities even in these smaller score ranges of less than 5% of the scale, and stays on top for all the bitrate values.

Results of the MS-SSIM tests on the "Crowd Run" sequence, seen in figure 21, are a bit puzzling. It is always possible that there has been a

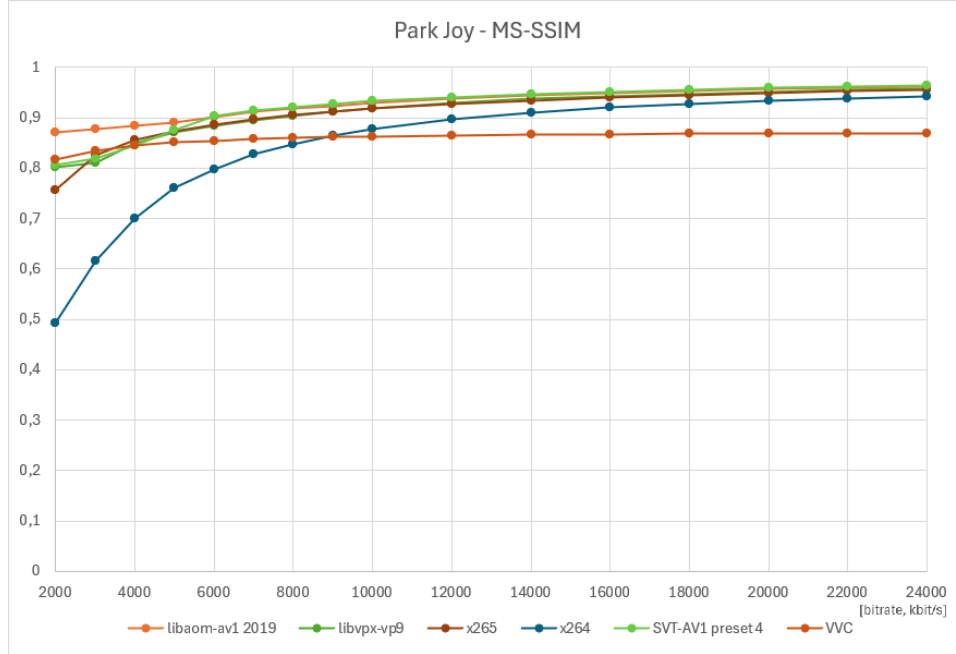
Figure 21: The results of the MS-SSIM tests performed on the "Crowd Run" sequence. Note: chart created by author using measured data, along with data retrieved from *Objektivní porovnání kvality video kódovacího formátu AV1* (P. Hovorka, 2021)



mistake made in testing, yet I wasn't able to find any reason why the two sequences shouldn't compare properly. VVC's score seems to barely change with bitrate, only slightly rising. It is somewhat level with the other results, but skewed by a bit down.

MS-SSIM results on the "Park Joy" sequence, seen in figure 22, are similar. VVC doesn't start of with a bad score, although at this point, the expectation would be for it to be on the top. It proceeds to very slowly improve with rising bitrate values, ending up with a relatively undesirable and unexpected score.

Figure 22: The results of the MS-SSIM tests performed on the "Park Joy" sequence. Note: chart created by author using measured data, along with data retrieved from *Objektivní porovnání kvality video kódovacího formátu AV1* (P. Hovorka, 2021)



5.2.3 PSNR-HVS-M - quality comparison

PSNR-HVS-M testing on "Crowd Run" and "Park Joy" yielded results that were certainly wrong, apparently due to a technical issue which I didn't end up being able to resolve. Only the testing on the "Sintel" sequence with the PSNR-HVS-M method yielded satisfactory results. Here, scores can go as high as 50 db, more being better,

The results of PSNR-HVS-M tests on the "Sintel" sequence are on display in figure 23. AV1 codecs, along with libvpx-vp9 and x265 show somewhat similar results within the lower bitrate values, however, they seem to diverge more as bitrate values rise. Excluding VVC, at higher bitrate values, the best scores are achieved by libvpx-vp9, with x265 closely following it. Doing slightly worse, libaom-av1's 2019 version and SVT-AV1 on its preset 4 follow. Again, VVC shows its clear superiority in this test as well, as it scores higher at all bitrates, with the greatest margins in the 4-12Mbit s⁻¹ range.

Figure 23: The results of the PSNR-HVS-M tests performed on the "Sintel" sequence. Note: chart created by author using measured data, along with data retrieved from *Objektivní porovnání kvality video kódovacího formátu AV1* (P. Hovorka, 2021)



5.3 Other Parameters

During encoding, other parameters were also measured, namely speed of encoding and peak ram usage - meaning the maximum ram the program has had allocated at one time throughout the run of the process.

5.3.1 Commands used

For measuring time, the linux "date" command was used. For measuring peak ram usage, the utility "/usr/bin/time" was used, a utility that can measure certain system parameters of processes, not to be confused with the "time" command. The -v argument makes the output "verbose", making it give more information. Its output is written to the `stderr` output stream, so `2>` has to be used to capture it.

All commands were put inside scripts, where they ran automatically, and logged their progress. All scripts were written by the author.

Time measurement:

```
START_TIME=$(date +%s)
$ENCODING_COMMAND
DURATION=$(( $(date +%s) - $START_TIME ))
```

RAM measurement:

```
/usr/bin/time -v $ENCODING_COMMAND 2>$$PROCESS_INFORMATION
```

Parameters:

`$START_TIME` - time at the start of the process, here stored as the number of seconds since "epoch time"

`$DURATION` - duration of the process run.

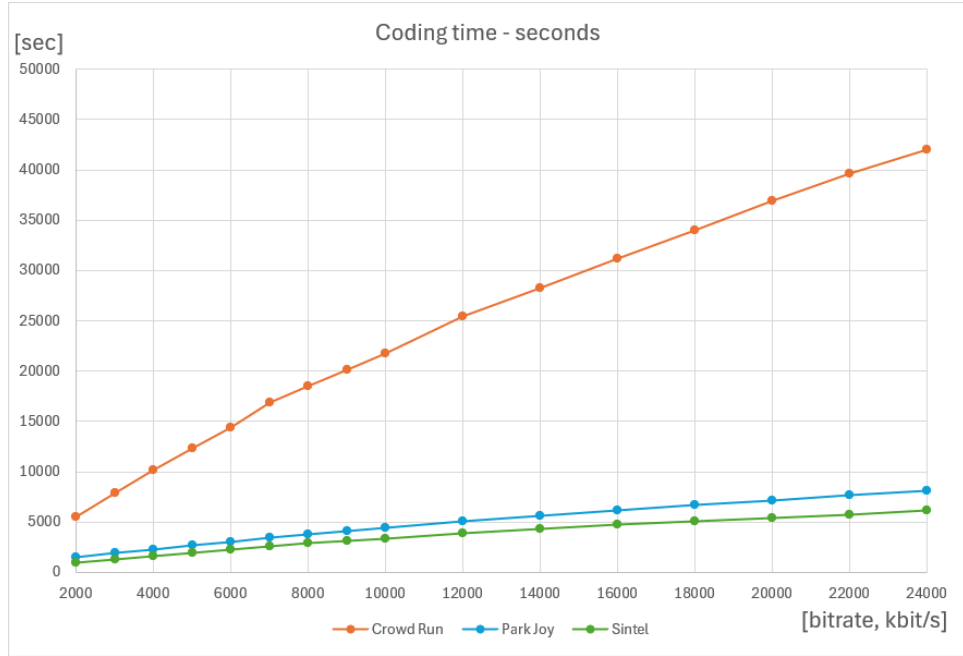
`$ENCODING_COMMAND` - command used to encode sequences with ffmpeg.

`$PROCESS_INFORMATION` - a temporary file storing process information, which is then filtered for the RAM usage result, and discarded.

5.3.2 Test Results

As seen on figure 24, encoding times rise approximately linearly with rising bitrate. It's worth reiterating that the processes always ran on all 6 available threads of the CPU, and all 3 sequences are 10 seconds, long, although the "Sintel" sequence has approximately half the frames, having a lower frame rate. A significant difference can be seen between the "slower" preset used on the "Crowd Run" sequence, and the "slow" preset used on the "Sintel" and "Park Joy" sequences.

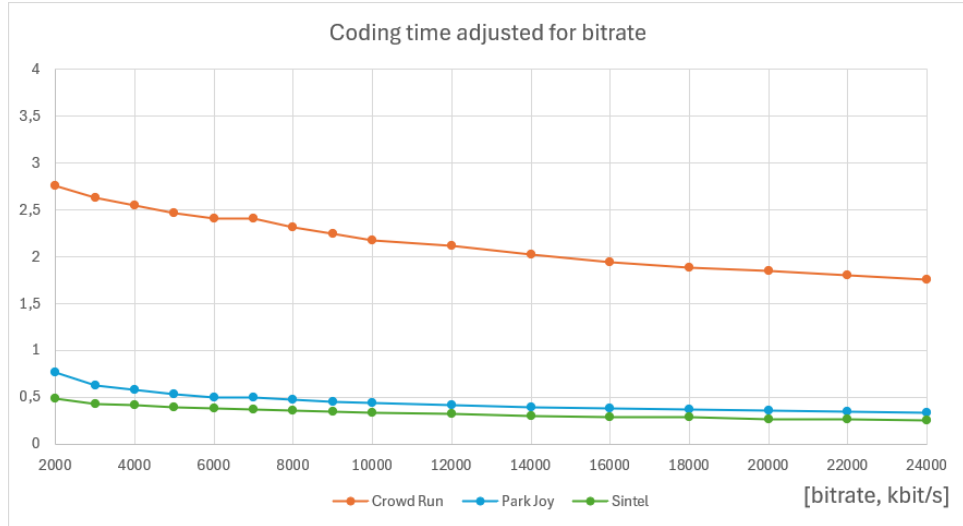
Figure 24: Coding times for coding processes. Note: chart created by author using measured data.



If we adjust the time values for the bitrate at which the encoding was performed, as seen in figure 25, a slight downward trend appears. All this means is that as we go up in bitrate, coding time increases more and more slowly. For example, if we look at time values for 2000 kbit s^{-1} and 4000 kbit s^{-1} , time spent encoding about doubles, just like the bitrate. On the other hand, if we look at the increase in encoding time from 4000 kbit s^{-1} to $16000 \text{ kbit s}^{-1}$, based on the first measurement, we could expect the time to quadruple, but the multiplier is much closer to $3\times$.

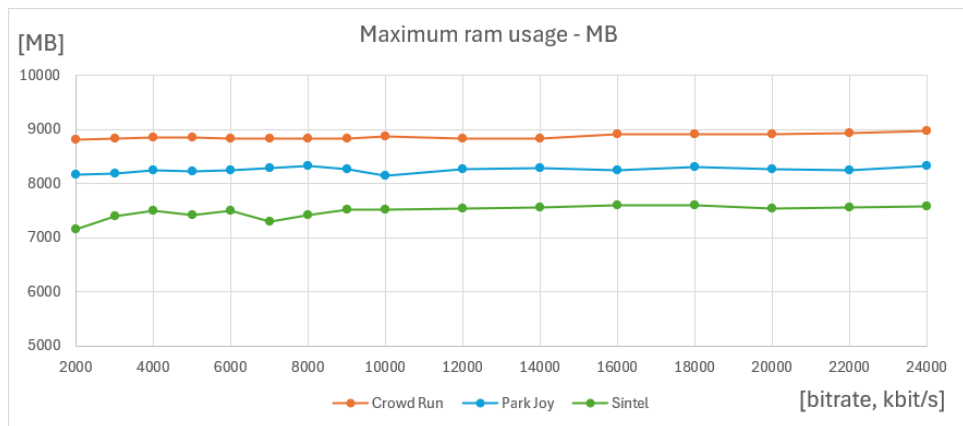
Lastly, measurements of peak RAM usage. Figure 26 shows that VVC encoding isn't exactly light on RAM usage. The most ram at one time was used by the encoding of the "Crowd Run" sequence, at the "slower" preset,

Figure 25: Coding time, adjusted for bitrate. Note: chart created by author using measured data.



each time around 9 GB of RAM. RAM peaked at lower values for the "Park Joy" sequence, around 8 GB at all bitrates. This can probably be quite safely assumed to be mostly caused by using the "slow" preset, since the sequence is in all parameters identical to the "Crowd Run" sequence, the only difference being its contents. The "Sintel" sequence peaked at the least amount of ram, around 7 GB for all bitrates. It ran on the same "slow" preset as the "Park Joy" sequence, but there is a slight difference in spatial frame size. For each frame, the "Sintel" sequence has about 13% less total pixels than "Crowd Run" and "Park Joy", quite well corresponding to the difference in peak RAM.

Figure 26: Peak ram usage for coding processes. Note: chart created by author using measured data.



Conclusion

Based on the data measured, given some outliers, it can overall be said that VVC brings a significant improvement to state-of-the-art video coding technology available today, and seems to be the best option in terms of quality. Although the format isn't royalty-free, information available seems to hint at this format slowly exiting its infancy, and moving into wider industrial use, having the backing of a mix of many major corporations, many of whose business relies on advances in multimedia technology like this.[63][22]

The utility of this thesis could be to inform potential users of this technology of some technical aspects, and help them choose the best option for their desired purpose, with respect to their requirements. Quality is among the, if not the most important aspect when choosing technical options for multimedia processing. Users might also be concerned with software requirements that come with VVC, this thesis should also give readers an idea about those aspects.

Of course, I'm aware of my work's limitations. Some areas for improvement are certainly in the testing section, especially in testing with the PSNR-HVS-M method, where I failed to resolve an issue with some of the sequences stopping me from properly testing them. Future works on this topic could also, for example, test different presets of the codec, and determine performance-to-quality compromises, or test encoding parameters on different numbers of CPU threads, as all of this information could also be relevant to potential users.

Terms

- **Aspect ratio** - Typically referring to video files or video transmission devices, an aspect ratio is the ratio between the width and height of the medium, in that order. Some common aspect ratios are 16:9, 4:3 or 9:16, for vertically aligned devices.[74]
- **Bitrate** - A Bitrate in context of multimedia is the amount of memory a multimedia stream takes up per a specified time of its run. Most commonly, bitrate is expressed in kbit s^{-1} or Mbit s^{-1} . [80]
- **Epoch time** - In the context of Unix-based systems, refers to the timestamp of 1.1.1970 at 00:00. Dates or times, especially those used by computer processing, are often expressed in terms of the duration since that timestamp, like in the case of calling the `date +%s` command on manjaro. [21]
- **HDMI** - High-Definition Multimedia Interface (HDMI) is a proprietary audio/video interface for transmitting compressed or uncompressed digital audio data and uncompressed video data from an HDMI-compliant source device, such as a display controller, to a compatible computer monitor, video projector, digital television, or digital audio device.[79]
- **HDMI** - High-Definition Multimedia Interface (HDMI) is a standard for transmitting audio and video in high quality from one device to another. It has been jointly developed by Hitachi, Masushita and sony to replace older connectors. It is among the most popular connectors of this type used today.[79]
- **Resolution** - Resolution, in the context of images/video frames, also referred to as pixel count, is the amount of pixels in an image/video, typically expressed as a multiplication of the number of rows and columns of pixels in an image, for example "1920 × 1080", or can assume the reader's knowledge of common resolutions and aspect ratios, and just express the height, such as "1080p", "p" standing for progressive, informing about the kind of scanning the format uses.[5]
- **YCbCr** - a family of color spaces where a color is represented by a luminance value - basically a brightness (represented by Y), and two chrominance values, Cb and Cr, named blue-difference and red-difference.[8]

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Appendices

Appendix A - quality comparison data

VMAF Test results for video sequences

Bitrate [kbit/s]	Sintel	Crowd Run	Park Joy
2000	94,065384	50,574789	52,209355
3000	98,130334	57,393269	58,037638
4000	99,394835	62,607868	62,4436
5000	99,681952	66,806997	66,095104
6000	99,725259	70,373479	68,983911
7000	99,74712	73,581844	71,807601
8000	99,765988	76,149255	73,983815
9000	99,773767	78,383086	75,915384
10000	99,785559	80,375116	77,5989
12000	99,797779	83,4171	80,508719
14000	99,800121	86,022967	82,926943
16000	99,809123	88,194189	85,132044
18000	99,809909	89,936661	86,819341
20000	99,809761	91,476253	88,354284
22000	99,819857	92,745663	89,870097
24000	99,819857	93,807982	91,061995

MS-SSIM Test results for video sequences

Bitrate [kbit/s]	Sintel	Crowd Run	Park Joy
2000	0,970532	0,827223	0,81665
3000	0,983002	0,847841	0,834418
4000	0,98917	0,858405	0,844273
5000	0,991726	0,864699	0,850494
6000	0,993298	0,868891	0,854457
7000	0,994516	0,87205	0,857974
8000	0,995228	0,874097	0,860011
9000	0,9957	0,875657	0,86166
10000	0,996133	0,876904	0,862855
12000	0,996723	0,878442	0,864868
14000	0,997022	0,879483	0,866154
16000	0,99727	0,880228	0,867254
18000	0,997434	0,880708	0,867877
20000	0,99757	0,881073	0,868364
22000	0,99769	0,881353	0,868824
24000	0,997798	0,881562	0,869166

PSNR-HVS-M Test results for video sequences

Bitrate [kbit/s]	Sintel
2000	35,642548
3000	38,421524
4000	40,492878
5000	41,798798
6000	42,860439
7000	43,853069
8000	44,513493
9000	45,097046
10000	45,654751
12000	46,54134
14000	47,097675
16000	47,577164
18000	47,992989
20000	48,296635
22000	48,610062
24000	48,8988

Appendix B - other parameters data

Encoding time for video sequences (seconds)

Bitrate [kbit/s]	Crowd Run	Park Joy	Sintel
2000	5521	1532	965
3000	7883	1881	1302
4000	10183	2312	1648
5000	12359	2689	1984
6000	14436	3027	2300
7000	16851	3454	2622
8000	18538	3768	2905
9000	20166	4115	3084
10000	21770	4375	3373
12000	25449	5044	3882
14000	28294	5572	4268
16000	31141	6161	4702
18000	34013	6647	5112
20000	36909	7139	5393
22000	39605	7699	5747
24000	42056	8133	6176

Peak ram usage during encoding for video sequences (KB)

Bitrate [kbit/s]	Crowd Run	Park Joy	Sintel
2000	9028820	8363160	7331600
3000	9049124	8374768	7588248
4000	9068992	8437232	7688016
5000	9059732	8433472	7605884
6000	9054076	8451324	7684860
7000	9049884	8487044	7470824
8000	9041168	8522228	7592768
9000	9050928	8477700	7710476
10000	9085080	8344016	7699948
12000	9040084	8476380	7731336
14000	9052904	8491720	7742584
16000	9129480	8437968	7777868
18000	9129572	8499772	7791496
20000	9125548	8477372	7732152
22000	9142548	8453000	7753516
24000	9183500	8521800	7772216